

Fortresses in Wolf and Lamb

26 September 2026

Summary

Both sides have a way of never losing, but they are of different kinds.

- **The lambs have true fortresses: shapes that cannot be broken whatever the wolves do.** None exists with 4 or 5 lambs; with 6 lambs there are exactly four (up to symmetry), with 7 lambs 141, with 8 lambs 1,179. They all follow one rule, the *distance-two rule*: a lamb can only be captured from a square two steps away in its row or column, so a lamb whose distance-two squares are all either off the board or occupied by lambs is untouchable. The catch is that lambs must move every turn, so a fortress also needs spare moves that do not open a hole. The 4-lamb square that satisfies the rule perfectly is nevertheless always lost, by zugzwang.
- **The wolves have no fortress shape.** Every wolf placement can be trapped for some lamb configuration. What the wolves have instead is a *principle*: stay apart and off the edges. With 9 lambs left, wolves that are not adjacent to each other and not on the edge lose in 0.9% of positions; wolves huddled on the edge lose in half of them. The reason is counting: a corner wolf is trapped by 2 lambs, an edge wolf by 3, a central wolf by 4, and three separate nets cost more lambs than the flock can spare.
- In the drawn starting line the two meet: the wolves eat down to a few lambs that have settled into a distance-two shape, the wolves are spread out, and nobody can make progress.

All statements below are read off the exact solution tables: for a shape, “never lost” means that over every placement of the three wolves and both sides to move, no position is a win for the other side.

The distance-two rule

A wolf captures by jumping over one empty square onto a lamb. So a lamb on square s can be captured along a line only if the neighbour of s on that line is empty and the square two steps away on the same side is on the board and reachable by a wolf. This gives:

A lamb is immune along a line if, on each side, the square two steps away is off the board or holds a lamb, or the adjacent square is occupied.

The second condition is fragile (the occupying piece may move), the first is stable as long as the protecting lamb stays. Squares two steps apart in a row or column have the same parity of row and of column, so the board splits into four *lattices*: the 9 squares with both coordinates odd, the 4 squares with both even, and two lattices of 6 with mixed parity. A set of lambs that fills a whole lattice is immune while it stands still:

even-even (4)	odd-even (6)	even-odd (6)
.	. L . L .	
. L . L .		L . L . L
.	. L . L .	
. L . L .		L . L . L
.	. L . L .	

Why the 4-lamb square is not a fortress

The four-square lattice is perfectly immune, yet the tables show it is lost in every position where the wolves are close. Lambs may not pass. Whatever lamb moves, it leaves its lattice square, loses the protection of its distance-two partners, and the lattice has no spare lamb to fill the gap; the wolves then win in a few moves. Example (wolf to move, wolves win in every line; also lost with the lambs to move, all 16 lamb moves losing):

```

W . . . .
. L . L .
. . W . .
. L . L .
. . . . W

```

With 4 lambs no shape at all is a fortress: one capture ends the game, and the best shape is still lost in 81 % of positions. With 5 lambs the best shape is lost in 36 %.

The 6-lamb fortresses

Six lambs is the smallest number that admits shapes which are never lost. There are exactly four up to symmetry:

A	B	C	D
. L . L .	. L . L .		
. L . L .		L . L . L	. L . L .
.	. L . L .		. L . L .
. L . L .	. L . L .	L . L . L	. L . L .
.			

C is the full even-odd lattice (6 lambs) and is drawn in all 524 positions. A, B and D are *double rails*: two columns one square in from the edges, three lambs each, with two of the three lambs stacked. Each rail protects the other at distance two horizontally; inside a rail the stacked pair protects itself by occupancy and the third lamb by distance two. A and B are drawn in 1,012 of 1,014 positions and won by the lambs in the other 2; nothing is ever lost.

What makes these six-lamb shapes work where the square failed is *tempo*. In shape A with the wolves probing as below, the lambs to move have 17 legal moves, 13 of which keep the draw:

```

. L . L .
. L . L .
W . W . .
. L . L .
. . . W .

```

Lamb to move: draw.
Drawing moves include (1,2)-(1,1), (1,4)-(1,3),
(2,2)-(2,1), (2,4)-(2,3), (4,2)-(4,1), (2,2)-(3,2), ...

A lamb steps sideways off its rail and back again; while it is away, its neighbours still cover the squares two steps from it, so the wolves gain nothing. The four losing moves are the ones

that put a lamb two steps from a wolf with an empty square between them, such as (4,4)-(3,4) next to the wolf on (3,3) or (4,2)-(5,2) in line with the wolf on (5,4).

Because the rule is local, the fortresses grow: with 7 lambs the extra lamb usually sits in a corner or on an edge square that the rails already protect, and with 8 lambs the most robust shapes put a full rail on the top edge:

7 lambs	8 lambs
. . . L L	L L L . L
. L . L .	. L . L .
.	
. L . L .	. L . L .
L	

Shapes that look just as tidy but break the rule are lost. The best five-lamb shape, a row of three with two supporters two rows below, is still lost in 36 % of positions; the best four-lamb shapes in 81 to 87 %.

The wolves: no shape, but a principle

For every placement of the three wolves, some lamb configurations trap them, even with only 7 lambs. So the wolves have no fortress in the strict sense. The tables do, however, show a strong pattern in *which* placements survive. Over all positions with 9 lambs:

adjacent wolf pairs	wolves on the edge	wolf wins	draws	lamb wins
0	0	54.3 %	44.8 %	0.9 %
0	1	36.5 %	59.2 %	4.3 %
0	2	18.6 %	68.1 %	13.3 %
0	3	7.7 %	62.1 %	30.2 %
1	3	9.5 %	50.6 %	39.9 %
2	3	12.8 %	36.2 %	50.9 %

Two effects, both large. Every wolf on the edge costs the wolves about a third of their winning chances and multiplies their losing chances. And wolves that touch each other are easier to trap than wolves that stay apart: with all three on the edge, going from no adjacent pair to two adjacent pairs raises the lamb win rate from 30 % to 51 %. A huddle can be closed with one net; three separated wolves need three nets, and the flock cannot afford them. The safest placements in the middle game are spread as far apart as the board allows:

7 lambs, 87% draws	8 lambs, 89% draws	9 lambs, 85% draws
W . . . W	 W	
.		
.	. . W . .	W . W . W
.		
W	W	

The count behind the principle: to immobilise a wolf the lambs must occupy all its empty neighbours *and* keep the square beyond each neighbour from giving a jump. In a corner that takes 2 lambs, on an edge 3, in the open 4, and the lambs also need tempo moves of their own. With three wolves far apart the bill exceeds what 7 to 9 lambs can pay, and the game is a draw.

How the two fortresses meet in the main line

From the start the only drawing wolf move is the central capture. The wolves then win material through double threats while the lambs give up lambs that cannot be saved and keep the ones that can. Best play by both sides reaches positions with a handful of lambs in a distance-two shape and three wolves spread across the board: the lambs are immune, the wolves are untrappable, and the 100-turn rule ends the game. One such ending, reached with the tables playing both sides:

W . W . .	
L . L . .	
L . . L .	Wolf to move, draw. Every lamb is covered by
W	a partner or an edge; every wolf has room.
.	

Practical advice

- **Lambs:** think in distance two. A lamb is safe if the squares two away in its row and column are lambs or off the board. Build rails a square in from the edge, keep pairs stacked, and always keep a harmless waiting move in reserve; if every move would open a hole, the shape is a zugzwang, not a fortress. Never trade the last spare lamb: six is the minimum that can hold.
- **Wolves:** never let two wolves touch, keep at least one wolf in the centre, and stay off the edges unless a capture forces it. A wolf that is tempted to the edge by a free lamb has often walked into the only net the lambs can afford.

How this was computed

`solver/fortress.js` lamb k groups every position with k lambs by the lamb shape (up to the eight symmetries of the board) and counts wolf wins, lamb wins and draws over all wolf placements and both sides to move; shapes with zero wolf wins are the fortresses. `solver/fortress.js` wolf k does the same by wolf placement, and `solver/wolfshape.js` k tabulates the value by the wolves' formation. The immunity of individual lambs was checked directly from the capture rule.